The documentation and process conversion measures necessary to comply with this document shall be completed by 29 March 2025.

INCH-POUND

MIL-PRF-19500/718A
w/AMENDMENT 1
29 October 2024
SUPERSEDING
MIL-PRF-19500/718A
22 March 2019

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, BIPOLAR TRANSIENT VOLTAGE SUPPRESSOR,
TYPES 1N6950 THROUGH 1N6986,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 5,000 watt, peak, pulse power, silicon, transient voltage suppressor diodes. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device as specified in MIL-PRF-19500.

1.2 Physical dimensions. The device package for the encapsulated device type is a surface mount UTK2 package in accordance with figure 1.

1.3 Maximum ratings. Maximum ratings are as shown in the electrical characteristics and ratings table herein (see 3.8), and as follows:

a. $P_{M(AV)}$ = 5.0 W (derate at 33.35 mW/°C above $T_A$ = +25°C) (see figure 2, and 6.6).

b. $P_{PP}$ = 5,000 W (see figure 3) at $t_p$ = 1 ms (see figure 4).

c. -65°C ≤ $T_j$ ≤ +175°C (ambient), -65°C ≤ $T_{STG}$ ≤ +175°C (ambient).

1.4 Primary electrical characteristics at $T_A$ = +25°C. Primary electrical characteristics are as shown in maximum test ratings herein (see 3.8).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

FSC 5961



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "6950" through "6986".

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contact.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

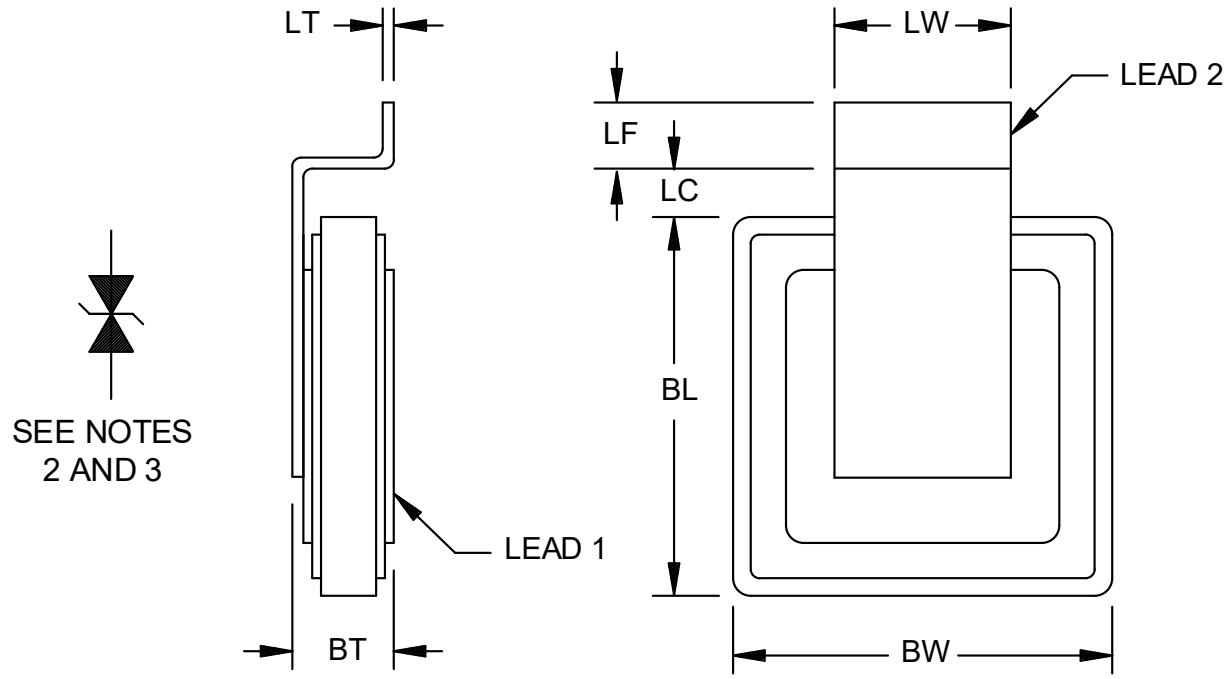
MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.




| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inch | Inch | mm | mm |
| | Min | Max | Min | Max |
| BW | .420 | .440 | 10.67 | 11.0 |
| BL | .420 | .440 | 10.67 | 11.0 |
| BT | .085 | .105 | 2.16 | 2.67 |
| LW | .195 | .205 | 4.95 | 5.21 |
| LT | .010 | .015 | 0.25 | 0.38 |
| LF | .070 | .080 | 1.78 | 2.03 |
| LC | .040 | .070 | 1.02 | 1.78 |

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Symbol for internal construction of bipolar transient suppressor.
3. The bottom of the case is lead 1 and the gull wing is lead 2.

FIGURE 1. Physical dimensions of a UTK2 surface mount package



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 herein.

3.4.1 Diode construction. These devices shall be constructed utilizing double plug construction with eutectic bonding between both sides of the silicon die and terminal pins (see MIL-PRF-19500). Metallurgical bond shall be in accordance with the requirements of category II in MIL-PRF-19500. The diode body is ceramic. All seals are eutectic solder. Strap material is a copper alloy or copper sandwich.

3.4.2 Lead finish. Unless otherwise specified, lead finish (pads, bottom pad and strap foot) shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750 and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I subgroup 2, table II, and table III.

3.8 Maximum test ratings. Test ratings for the devices specified herein shall be as shown in table IV.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.1.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-19500, and as specified herein except lot accumulation shall be 3-months in lieu of 6-weeks.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein shall be performed on the first inspection lot to this revision to maintain qualification.

4.3 Screening (quality levels JANS, JANTX, and JANTXV only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurements for quality level JANS | Measurements for quality levels JANTX and JANTXV |
|---|---|---|
| 3 | $T_{(high)}$ = +175°C | $T_{(high)}$ = +175°C |
| 9, 10, 11 | Not applicable | Not applicable |
| 12 | See 4.5.1 | See 4.5.1 |
| 13 | Interim electrical, delta, and table 1, subgroups 2 and 3, electrical parameters herein not applicable for this screen (performed in screen 12). | Interim electrical, delta, and table 1, subgroups 2 and 3, electrical parameters herein not applicable for this screen (performed in screen 12). |

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Delta measurement shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 1056<br>1051 | -55°C to 100°C, 25 cycles, n = 22, c = 0.<br>-55°C to +175°C, 100 cycles, n = 22, c = 0. Surge not applicable. |
| B4 | | Not applicable. |
| B5 | 1027 | Condition for accelerated steady-state operation life are as follows: See 4.5.2, $T_A$ = +100°C (min); $T_J$ = +225°C (min); t = 168 hours minimum. |
| B6 | | Not applicable. |

NOTE: All electrical measurements and tests shall be performed twice, once in each direction.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV (table E-VIB of MIL-PRF-19500).

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 1056 | -55°C to 100°C, 10 cycles, n = 22, c = 0. |
| | 1051 | -55°C to 175°C, 25 cycles, n = 22, c = 0. Surge not applicable. |
| B3 | 1027 | See 4.5.1, 1 ms pulse only (see 4.5.2). |
| B5 | | Not applicable |

NOTE: All electrical measurements and tests shall be performed twice, once in each direction.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Delta measurements shall be in accordance with the applicable steps of table III herein.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Lead tension: Test condition A; weight = 5 pounds (2.27 kg); t = 15 ±3 s. |
| C3 | | Not applicable |
| C5 | | Not applicable |
| C6 | 1026 | See 4.5.1, (see 4.5.2). |
| C7 | 1018 | |
| C8 | | Conditions for temperature coefficient of breakdown voltage are as follows: $I_{(BR)}$ = column 3 of table IV, $T_1$ = +25°C ±3°C, $T_2$ = $T_1$ +100°C; n = 22, c = 0. |
| C9 | | Conditions for maximum peak pulse current are as follows: See 4.5.3.a, (20 µs pulse only) 10 pulses; n = 22, c = 0. |

NOTE: All electrical measurements and tests shall be performed twice, once in each direction.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. See table III for delta measurements when applicable.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

4.5.1 Power burn-in (HTRB) and steady-state operation life test conditions. For the purposes of this test, the direction in which the device is first pulsed shall be considered polarity A and the reverse direction polarity B.

4.5.1.1 Screening and groups B and E procedure. The test conditions and order of events shall be as follows:

a. Pulse in accordance with 4.5.3.b, in polarity A 10 times for screening and group B and 50 times for group C at $T_A$ = +25°C.

b. Pulse in accordance with 4.5.3.b, in polarity B 10 times for screening and group B and 50 times for group C at $T_A$ = +25°C.

c. Read $I_D$ in polarities A and B at $T_A$ = +25C, remove defective devices and record the number of failures.

d. Apply the working peak reverse voltage ($V_{WM}$) (column 4 of table IV) at polarity A at $T_A$ = +150°C as follows:

(1) For 48 hours (JANTX and JANTXV) and 120 hours (JANS) for the screening test.

(2) For 170 hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

(3) For 500 hours for group C steady-state operation life test.

e. Read $I_D$ in polarity A at $T_A$ = +25°C. Devices with $\Delta I_D$ > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 µA dc, whichever is greater shall be considered defective. Remove defective devices and record the number of failures. 1/

f. Apply the working peak reverse voltage ($V_{WM}$) (column 4 of table IV) at polarity B at $T_A$ = +150°C as follows:

(1) For 48 hours (JANTX and JANTXV) and 120 hours (JANS) for the screening test.

(2) For 170 hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

(3) For 500 hours for group C steady-state operation life test.

g. Read $I_D$ in polarity B at $T_A$ = +25°C. Devices with $\Delta I_D$ > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 µA dc, whichever is greater shall be considered defective. Remove defective devices and record the number of failures.

h. Read $V_{(BR)}$ in polarities A and B at $T_A$ = +25°C. Devices with $\Delta V_{(BR)}$ > 2 percent (±5 percent for steady-state operation life) of the initial reading shall be considered defective. Remove defective devices and record the number of failures.

i. Read $I_D$ in polarity A at $T_A$ = +25°C, remove defective devices and record the number of failures.

4.5.1.2 Groups C steady-state operation life test (alternate procedure). When the group B, 340-hour life test is continued on test to 1,000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.5.1.1 with the following exceptions:

a. In 4.5.1.1, steps a and b shall be moved and performed following step g.

b. In 4.5.1.1, steps e and g shall be repeated after step a and b are performed and before step i is completed (step i may be omitted when this procedure is used.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

4.5.2 Accelerated steady-state operation life (JANS). This test shall be conducted with the devices subjected to the breakdown current specified in column 10 of table IV in opposite polarities for 84 +8, -4 hours in each polarity. At the beginning of the test and at the end of each time period, the devices shall be temperature stabilized at $T_A$ = +25°C) and subjected to pulse conditions at the rate of one pulse per minute (max) for 10 pulses in accordance with 4.5.3 as specified.

4.5.3 Maximum peak pulse current ($I_{pp}$). The peak pulse currents specified in column 7 of table IV shall be applied while simultaneously maintaining a bias voltage of not less than the applicable voltage specified in column 4 of table IV, in the same polarity as the peak pulse current. The peak pulse current shall be applied with a current vs time waveform as follows (1 pulse per minute maximum):

a. Pulse current shall reach 100 percent of $I_{PP}$ at t ≤ 8 µs and decay to 50 percent of $I_{PP}$ at t ≥ 20 µs for $t_p$ = 20 µs (see figure 5).

b. Pulse current shall reach 100 percent of $I_{PP}$ at t ≤ 10 µs and decay to 50 percent of $I_{PP}$ at t ≥ 1 ms for $t_p$ = 1 ms (see figure 4).

4.5.4 Clamping voltage. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent. The peak clamping voltage as specified in column 6 of table IV shall be applicable to the 1 ms pulse of 4.5.3.b only.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit 2/ | Limit 2/ | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 3/ | | | | | | |
| Standby current | 4016 | Condition A or B, $V_R = V_{WM}$ (column 4 of table IV) | $I_D$ | | Column 5 | µA dc |
| Breakdown voltage | 4022 | $I_{(BR)}$ = column 3 of table IV, $t_p \leq 300$ ms, duty cycle ≤ 2 percent | $V_{(BR)}$ | Column 2 | Column 2 | V dc |
| Subgroup 3 3/ | | | | | | |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Minimum breakdown voltage | 4022 | $I_{(BR)}$ = column 3 of table IV, $t_p \leq 300$ ms, duty cycle ≤ 2 percent | $V_{(BR)}$ | Column 9 | | V dc |
| Subgroup 4 3/ | | | | | | |
| Clamping voltage maximum (pulsed) | | See 4.5.4, $t_p$ = 10/1000 µs (see 4.5.3.b), $I_{PP}$ = column 7 of table IV | $V_C$ | | Column 6 | V (pk) |
| Subgroups 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table IV.
3/ All electrical testing shall be performed twice, once in each direction.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

TABLE II. Group E inspection (all quality levels) for qualification only.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices, c = 0 |
| Thermal shock | 1056 | 500 cycles, condition A, 0°C to +100°C. | |
| Temperature cycling | 1051 | 500 cycles, condition C, -65°C to +175°C. | |
| Electrical measurements | | See table III, steps 1, 2, and 5. | |
| Subgroup 2 | | | 22 devices, c = 0 |
| Life test | | 1,000 hours, see 4.5.1. | |
| Electrical measurements | | See table III, steps 1, 2, 3, 4, and 5. | |
| Subgroup 4 | | | |
| Thermal impedance curves | | Each supplier shall submit their (typical) design thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | |
| Peak pulse current | | See 4.5.3. $I_{PP}$ shall be characterized by the supplier and this data shall be available to the government. Test shall be performed on highest and lowest voltage in each structurally identical grouping. Test until failure occurs on all devices within the chosen sample size to characterize each voltage, or until the rated $I_{PP}$ is exceeded by 30%. | |
| Electrical measurements | | See table III, steps 1 and 2. | |
| Subgroup 7 | | | 45 devices, c = 0 |
| Resistance to soldering heat | 2031 | 1 cycle | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

TABLE III. Groups B and C electrical and delta measurements. 1/ 2/ 3/ 4/

| Step | Inspection | Method | Conditions | Symbol | Limit 5/ Min | Limit 5/ Max | Unit |
|---|---|---|---|---|---|---|---|
| 1. | Standby current | 4016 | Condition A or B; $V_R = V_{WM}$ (column 4 of table IV) | $I_D$ | | Column 5 | µA dc |
| 2. | Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table IV | $V_{(BR)}$ | Column 2 | Column 2 | V dc |
| 3. | Standby current | 4016 | Condition A or B; $V_R = V_{WM}$ (column 4 of table IV) | $\Delta I_D$ 6/ | 100 percent of initial reading or 20 percent of column 5 of table IV, whichever is greater. | | |
| 4. | Breakdown voltage | 4022 | $I_{(BR)}$ = column 3 of table IV; $t_p \leq 300$ ms, duty cycle ≤ 2 percent | $\Delta V_{(BR)}$ 6/ | | ±5 percent of initial value | |
| 5. | Clamping voltage | | $t_p$ = 10/1000 µs (see 4.5.3.b); $I_{PP}$ = column 7 of table IV | $V_C$ | | Column 6 | V (pk) |

1/ All electrical testing shall be performed twice, once in each direction.
2/ The electrical measurements taken during group B inspection for quality level JANS (table E-VIA of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 3, the measurements of steps 1, 2 and 5 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 5, the measurements of 4.5.1 shall also be taken.
3/ The electrical measurements taken during group B inspection for quality levels JAN, JANTX, and JANTXV (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 3, the measurements of 4.5.1 shall also be taken.
   c. In addition to the measurements specified for subgroup 6, the measurements of steps 1, 2, 3 and 4 of this table shall also be taken.
4/ The electrical measurements taken during group C inspection for all quality levels (table E-VII of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroups 2 and 9, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 6, the measurements of 4.5.1 shall also be taken.
5/ Column references are to table IV.
6/ Devices which exceed the table I limits for this test shall not be accepted.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

TABLE IV. Characteristics and ratings.

| Col 1 | Col 2 | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | | Col 8 | Col 9 | Col 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Type | Breakdown voltage $V_{(BR)}$ at $I_{(BR)}$ | Test current $I_{(BR)}$ | Working peak reverse voltage $V_{WM}$ | Maximum standby current $I_D$ | Maximum clamping voltage $V_C$ at $I_{PP}$ | Maximum peak pulse current ($I_{PP}$) | | Maximum temperature coefficient of $V_{(BR)}$ $\alpha V_{(BR)}$ | Minimum breakdown voltage at $I_{(BR)}$ $T_A$ = -55°C | Breakdown current $I_{(BR)}$ Maximum dc current $T_A$ = +25°C |
| | | | | | | $t_p$ = 20 µs $t_r$ = 8 µs | $t_p$ = 1 ms $t_r$ = 10 µs | | | |
| | Min V dc | mA dc | V (pk) | µA dc | V (pk) | A (pk) | A (pk) | α/°C | V dc | mA dc |
| 1N6950 | 7.13 | 10 | 6.0 | 1000 | 11.3 | 2500 | 440 | 0.061 | 6.66 | 625.0 |
| 1N6951 | 7.79 | 10 | 7.0 | 500 | 12.1 | 2333 | 413 | 0.065 | 7.24 | 575.0 |
| 1N6952 | 8.65 | 10 | 7.5 | 200 | 13.4 | 2100 | 373 | 0.068 | 8.01 | 520.0 |
| 1N6953 | 9.50 | 1 | 8.5 | 50 | 14.5 | 1950 | 343 | 0.073 | 8.75 | 475.0 |
| 1N6954 | 10.50 | 1 | 9.0 | 10 | 15.6 | 1817 | 320 | 0.075 | 9.65 | 430.0 |
| 1N6955 | 11.40 | 1 | 10.0 | 5 | 16.7 | 1700 | 300 | 0.078 | 10.40 | 365.0 |
| 1N6956 | 12.40 | 1 | 11.0 | 5 | 18.2 | 1550 | 273 | 0.081 | 11.30 | 350.0 |
| 1N6957 | 14.30 | 1 | 12.0 | 5 | 21.2 | 1333 | 237 | 0.084 | 13.00 | 315.0 |
| 1N6958 | 15.20 | 1 | 13.0 | 5 | 22.5 | 1250 | 223 | 0.086 | 13.70 | 295.0 |
| 1N6959 | 17.10 | 1 | 15.0 | 5 | 25.2 | 1117 | 198 | 0.088 | 15.40 | 265.0 |
| 1N6960 | 19.00 | 1 | 17.0 | 5 | 27.7 | 1018 | 180 | 0.090 | 17.10 | 235.0 |
| 1N6961 | 20.90 | 1 | 18.0 | 5 | 30.6 | 917 | 163 | 0.092 | 18.80 | 215.0 |
| 1N6962 | 22.80 | 1 | 20.0 | 5 | 33.2 | 850 | 150 | 0.094 | 20.50 | 195.0 |
| 1N6963 | 25.70 | 1 | 22.0 | 5 | 37.5 | 750 | 133 | 0.096 | 23.00 | 175.0 |
| 1N6964 | 28.50 | 1 | 25.0 | 5 | 41.4 | 683 | 120 | 0.097 | 25.50 | 155.0 |
| 1N6965 | 31.40 | 1 | 28.0 | 5 | 45.7 | 617 | 110 | 0.098 | 29.00 | 140.0 |
| 1N6966 | 34.20 | 1 | 30.0 | 5 | 49.9 | 567 | 100 | 0.098 | 30.50 | 130.0 |
| 1N6967 | 37.10 | 1 | 33.0 | 5 | 53.9 | 518 | 93.3 | 0.100 | 33.10 | 120.0 |
| 1N6968 | 40.90 | 1 | 36.0 | 5 | 59.3 | 483 | 84.3 | 0.101 | 36.40 | 110.0 |
| 1N6969 | 44.70 | 1 | 40.0 | 5 | 64.8 | 433 | 77.3 | 0.101 | 39.80 | 100.0 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

TABLE IV. Characteristics and ratings − Continued.

| Col 1 | Col 2 | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | | Col 8 | Col 9 | Col 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Type | Breakdown voltage $V_{(BR)}$ at $I_{(BR)}$ | Test current $I_{(BR)}$ | Working peak reverse voltage $V_{WM}$ | Maximum standby current $I_D$ | Maximum clamping voltage $V_C$ at $I_{PP}$ | Maximum peak pulse current ($I_{PP}$) | | Maximum temperature coefficient of $V_{(BR)}$ $\alpha V_{(BR)}$ | Minimum breakdown voltage at $I_{(BR)}$ $T_A$ = -55°C | Breakdown current $I_{(BR)}$ Maximum dc current $T_A$ = +25°C |
| | | | | | | $t_p$ = 20 µs $t_r$ = 8 µs | $t_p$ = 1 ms $t_r$ = 10 µs | | | |
| | Min V dc | mA dc | V (pk) | µA dc | V (pk) | A (pk) | A (pk) | α/°C | V | mA dc |
| 1N6970 | 48.50 | 1 | 43.0 | 5 | 70.1 | 400 | 71.3 | 0.102 | 43.10 | 90.0 |
| 1N6971 | 53.20 | 1 | 47.0 | 5 | 77.0 | 367 | 65 | 0.103 | 47.30 | 85.0 |
| 1N6972 | 58.90 | 1 | 53.0 | 5 | 85.0 | 333 | 59 | 0.104 | 52.30 | 75.0 |
| 1N6973 | 64.60 | 1 | 58.0 | 5 | 92.0 | 300 | 54.3 | 0.104 | 57.30 | 70.0 |
| 1N6974 | 71.30 | 1 | 64.0 | 5 | 103.0 | 273 | 48.7 | 0.105 | 63.20 | 60.0 |
| 1N6975 | 77.90 | 1 | 70.0 | 5 | 113.0 | 250 | 44.3 | 0.105 | 69.00 | 55.0 |
| 1N6976 | 86.50 | 1 | 75.0 | 5 | 125.0 | 227 | 40 | 0.106 | 76.50 | 50.0 |
| 1N6977 | 95.00 | 1 | 82.0 | 5 | 137.0 | 207 | 36.7 | 0.106 | 84.10 | 47.5 |
| 1N6978 | 105.00 | 1 | 94.0 | 5 | 152.0 | 183 | 33 | 0.107 | 92.80 | 42.5 |
| 1N6979 | 114.00 | 1 | 100.0 | 5 | 168.0 | 167 | 29.7 | 0.107 | 100.00 | 37.5 |
| 1N6980 | 124.00 | 1 | 110.0 | 5 | 182.0 | 157 | 27.3 | 0.107 | 109.00 | 35.0 |
| 1N6981 | 143.00 | 1 | 128.0 | 5 | 213.0 | 133 | 23.3 | 0.108 | 126.00 | 30.0 |
| 1N6982 | 162.00 | 1 | 145.0 | 5 | 245.0 | 120 | 20.3 | 0.108 | 143.00 | 27.5 |
| 1N6983 | 171.00 | 1 | 150.0 | 5 | 261.0 | 113 | 19 | 0.108 | 151.00 | 26.0 |
| 1N6984 | 181.00 | 1 | 160.0 | 5 | 278.0 | 107 | 18 | 0.108 | 157.00 | 25.0 |
| 1N6985 | 190.00 | 1 | 170.0 | 5 | 294.0 | 100 | 17 | 0.108 | 169.00 | 23.5 |
| 1N6986 | 209.00 | 1 | 185.0 | 5 | 328.0 | 83 | 15.3 | 0.108 | 184.00 | 21.5 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

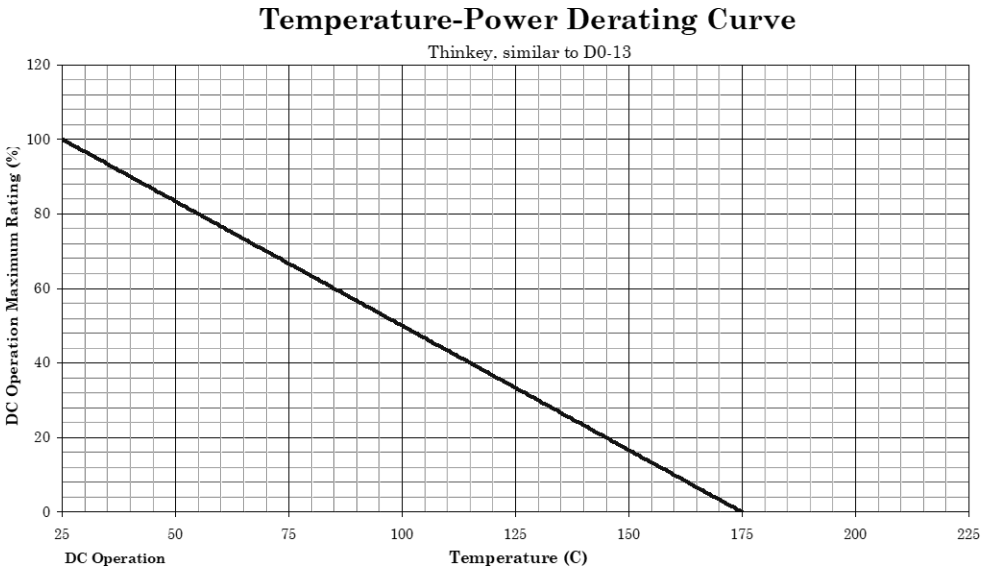



FIGURE 2. Derating curve.

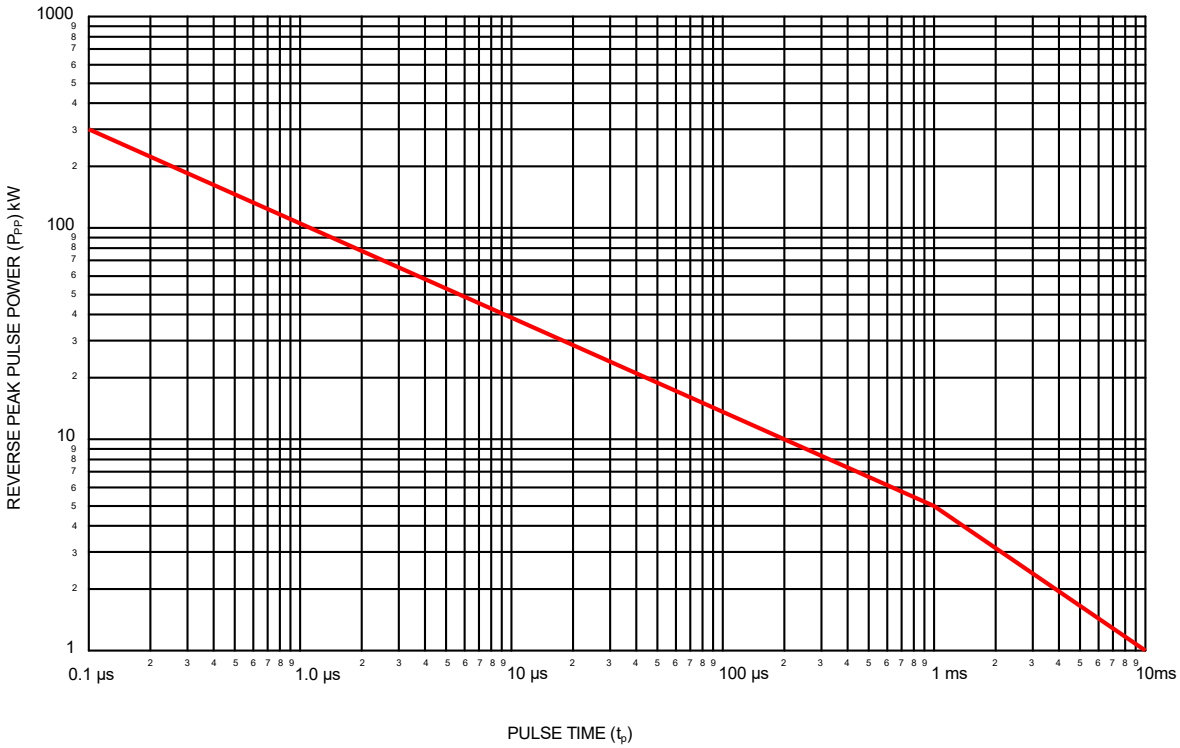



FIGURE 3. Peak pulse power versus pulse time.

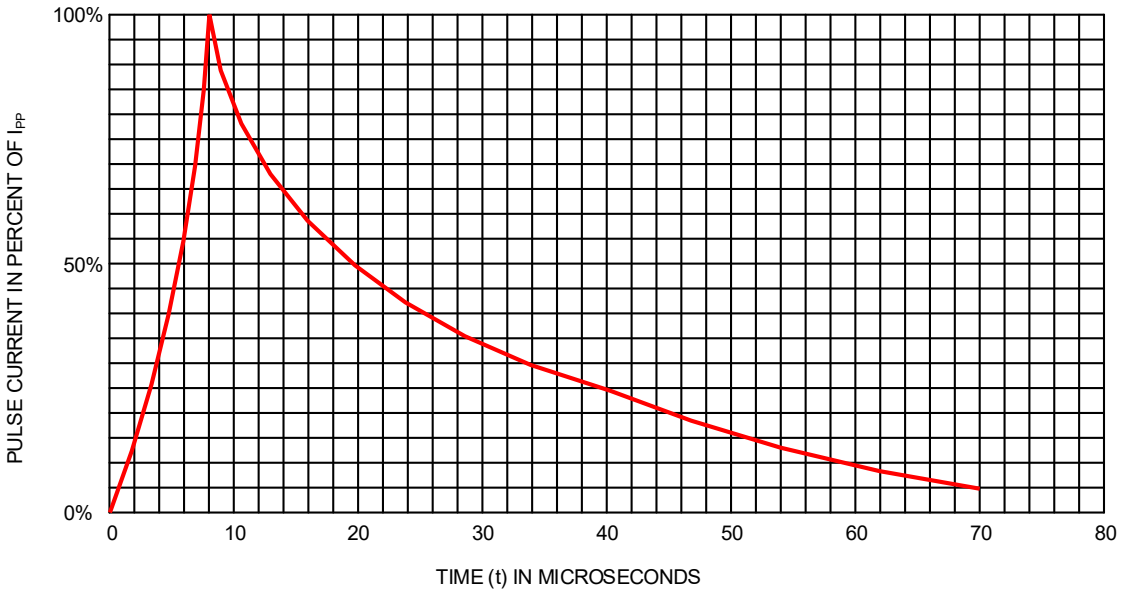


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.




NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of $I_{PP}$. (Rise time to 100 percent of $I_{PP}$ = 10 µs).

FIGURE 4. Current impulse waveform.




NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of $I_{PP}$. (Rise time to 100 percent of $I_{PP}$ = 8 µs).

FIGURE 5. Current impulse waveform.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

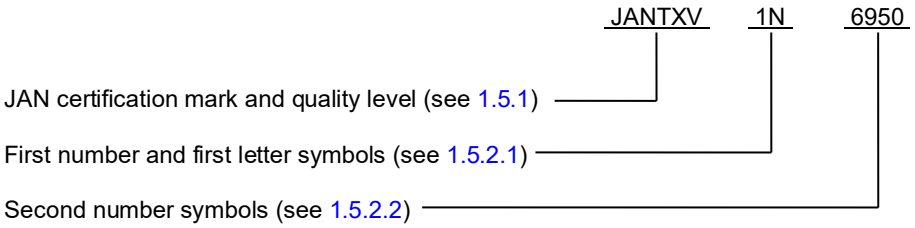
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.2).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.






Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

| PINs for types 1N6950 through 1N6986. | | | |
|---|---|---|---|
| JAN1N6950 | JANTX1N6950 | JANTXV1N6950 | JANS1N6950 |
| JAN1N6951 | JANTX1N6951 | JANTXV1N6951 | JANS1N6951 |
| JAN1N6952 | JANTX1N6952 | JANTXV1N6952 | JANS1N6952 |
| JAN1N6953 | JANTX1N6953 | JANTXV1N6953 | JANS1N6953 |
| JAN1N6954 | JANTX1N6954 | JANTXV1N6954 | JANS1N6954 |
| JAN1N6955 | JANTX1N6955 | JANTXV1N6955 | JANS1N6955 |
| JAN1N6956 | JANTX1N6956 | JANTXV1N6956 | JANS1N6956 |
| JAN1N6957 | JANTX1N6957 | JANTXV1N6957 | JANS1N6957 |
| JAN1N6958 | JANTX1N6958 | JANTXV1N6958 | JANS1N6958 |
| JAN1N6959 | JANTX1N6959 | JANTXV1N6959 | JANS1N6959 |
| JAN1N6960 | JANTX1N6960 | JANTXV1N6960 | JANS1N6960 |
| JAN1N6961 | JANTX1N6961 | JANTXV1N6961 | JANS1N6961 |
| JAN1N6962 | JANTX1N6962 | JANTXV1N6962 | JANS1N6962 |
| JAN1N6963 | JANTX1N6963 | JANTXV1N6963 | JANS1N6963 |
| JAN1N6964 | JANTX1N6964 | JANTXV1N6964 | JANS1N6964 |
| JAN1N6965 | JANTX1N6965 | JANTXV1N6965 | JANS1N6965 |
| JAN1N6966 | JANTX1N6966 | JANTXV1N6966 | JANS1N6966 |
| JAN1N6967 | JANTX1N6967 | JANTXV1N6967 | JANS1N6967 |
| JAN1N6968 | JANTX1N6968 | JANTXV1N6968 | JANS1N6968 |
| JAN1N6969 | JANTX1N6969 | JANTXV1N6969 | JANS1N6969 |
| JAN1N6970 | JANTX1N6970 | JANTXV1N6970 | JANS1N6970 |
| JAN1N6971 | JANTX1N6971 | JANTXV1N6971 | JANS1N6971 |
| JAN1N6972 | JANTX1N6972 | JANTXV1N6972 | JANS1N6972 |
| JAN1N6973 | JANTX1N6973 | JANTXV1N6973 | JANS1N6973 |
| JAN1N6974 | JANTX1N6974 | JANTXV1N6974 | JANS1N6974 |
| JAN1N6975 | JANTX1N6975 | JANTXV1N6975 | JANS1N6975 |
| JAN1N6976 | JANTX1N6976 | JANTXV1N6976 | JANS1N6976 |
| JAN1N6977 | JANTX1N6977 | JANTXV1N6977 | JANS1N6977 |
| JAN1N6978 | JANTX1N6978 | JANTXV1N6978 | JANS1N6978 |
| JAN1N6979 | JANTX1N6979 | JANTXV1N6979 | JANS1N6979 |
| JAN1N6980 | JANTX1N6980 | JANTXV1N6980 | JANS1N6980 |
| JAN1N6981 | JANTX1N6981 | JANTXV1N6981 | JANS1N6981 |
| JAN1N6982 | JANTX1N6982 | JANTXV1N6982 | JANS1N6982 |
| JAN1N6983 | JANTX1N6983 | JANTXV1N6983 | JANS1N6983 |
| JAN1N6984 | JANTX1N6984 | JANTXV1N6984 | JANS1N6984 |
| JAN1N6985 | JANTX1N6985 | JANTXV1N6985 | JANS1N6985 |
| JAN1N6986 | JANTX1N6986 | JANTXV1N6986 | JANS1N6986 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.

6.6 Steady state power rating. This rating is not relevant for most applications.

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army − CR
Navy − SH
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − MI
Air Force − 19

Preparing activity:
DLA − CC

(Project 5961−2024−064)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:09Z
Check the source to verify that this is the current version before use.